Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/21/2004
PI Name:	Ladd, Anthony J.C. Ph.D.		
Project Title:	Dynamics and Structures in Colloidal Suspensions far from Equilibrium		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ladd@che.ufl.edu	**Fax:**	FY 352 392-9513
PI Organization Type:	UNIVERSITY	**Phone:**	352 392-6509
Organization Name:	University of Florida		
PI Address 1:	P.O. Box 116005		
PI Address 2:			
PI Web Page:	http://ladd.che.ufl.edu		
City:	Gainesville	**State:**	FL
Zip Code:	32611-6005	**Congressional District:**	6
Comments:	PI has completed submission (kw 11/30)		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:			
Flight Assignment:	Note: Original end date of 10/30/2007 changed to 9/30/2006.		
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We will carry out a coordinated sequence of numerical simulations and laboratory experiments to address fundamental questions related to the dynamics of batch sedimentation, fluidized beds, and centrifugal flows. In some instances the crucial scientific issues can be addressed by ground-based experiments, supported by simulations. However, there are other questions that can only be answered in a microgravity environment. We propose to simulate suspension flows in the absence of gravity to obtain partial answers to these questions in the near term, and also to guide a future flight experiment. The proposed work follows on from current NASA-supported investigations, which will end in November 2003. The objective of this work is to continue our ongoing efforts to develop a better understanding of gas fluidized systems by extending the current limits of laboratory experiments and numerical simulation. We use experiments and simulations synergistically; thus we not only use laboratory experiments to validate the results of numerical simulations,

Task Description:	but we also use simulations to clarify aspects of laboratory experiments, such as polydispersity, that are difficult to control experimentally. Numerical simulations will also be used to provide preliminary answers to questions that can only be fully investigated in a microgravity environment. Specifically we will attempt to answer three fundamental questions related to particle-fluid flows, motivated in part by what was discovered in the past 3 years research. The first question is to understand the origin of the observed correlation length in settling suspensions and decide if it is universal, applying also to fluidized suspensions. The second issue is to discover if particle collisions in a low-Reynolds number fluidized bed lead to important changes in suspension microstructure at finite Stokes numbers, and to investigate whether such microstructural changes are related to the onset of instability in gas-fluidized suspensions at low Stokes numbers. Numerical simulations will be used to uncover the fundamental physics underlying the effects of small amounts of particle inertia, which under normal conditions is masked by the effects of the earth's gravitational field. Although it is possible to study liquid-fluidized suspensions at finite-Stokes numbers in the laboratory, the parameter ranges that characterize gas-fluidized flows can only be achieved in a reduced gravity environment. Finally we will examine a flow with a non-uniform external force, namely a rotating suspension flow. In this case the centrifugal force produces a non-uniform body force on the particles, leading to a surprising segregation of particles along the rotation axis into bands of high and low concentration. In this case it is essential to remove gravity altogether, so that the effects of the centrifugal force can be studied in isolation. Moreover, the bands of high and low concentration may develop differently in a microgravity environment than on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Lattice-Boltzmann simulations have been used to investigate low-Reynolds-number settling of monodisperse and polydisperse suspensions. We confirm the discovery that particle velocity fluctuations are strongly suppressed by no-slip walls at the top and bottom of the system, even in regions distant from the boundaries. We also show that a monodisperse suspension develops a strongly anisotropic long-range microstructure during the settling process, with vanishing density fluctuations in the horizontal plane. We find no numerical evidence that the particle concentration in the bulk is stratified; diffusive spreading of the suspension-supernatent interface is suppressed by hindered settling, as would be expected in moderately concentrated suspensions. Long-range correlations in particle density fluctuations are destroyed by polydispersity in particle size, and in this case density fluctuations are finite at all length scales and in all directions. However, in polydisperse suspensions there is significant stratification, due to differential settling rather than interface diffusion, which provides an alternate mechanism for screening the hydrodynamic interactions. It is possible that this is the dominant mechanism for hydrodynamic screening in several laboratory experiments.
Bibliography Type:	Description: (Last Updated: 03/15/2007)
Abstracts for Journals and Proceedings	N.-Q. Nguyen and A.J.C. Ladd. "Sedimentation of hard-sphere suspensions at low Reynolds number." Journal of Fluid Mechanics In Press , Jan-2005